

Homework 1

Due September 28, 2026

Instructions

- Explain your reasoning. A correct answer without supporting work may receive little credit.
- You may use the hints freely.
- The four required problems total 100 points.
- The FreeFEM++ problem is optional and is worth up to 15 bonus points.

Problem 1: A holomorphic flow (20 points)

Consider the complex potential

$$F(z) = \frac{z^3}{3},$$

and recall that its associated velocity field is $V = \overline{F'(z)}$.

1. Write $V = (u, v)$ explicitly in terms of x and y . (4 points)
2. Verify directly that V is divergence-free and curl-free. (5 points)
3. Find an equation for the streamlines and sketch the streamline $\text{Im } F = 0$ together with several nearby streamlines. Indicate the direction of the flow. (6 points)
4. Find every stagnation point and express the speed $|V|$ in terms of $|z|$. (2 points)
5. Let C be the vertical segment from $A = (0, -1)$ to $B = (0, 1)$, oriented upward. Compute the flux across C both directly and from $\text{Im } F(B) - \text{Im } F(A)$. (3 points)

Hints.

- If $F' = a + ib$, then $V = (a, -b)$.
- Expand $(x + iy)^2$ and $(x + iy)^3$ before taking real and imaginary parts.
- Streamlines are the level curves $\text{Im } F = c$.
- On an oriented curve, the convention from the notes is $n \, ds = (dy, -dx)$.
- Your two flux calculations should have the same sign as well as the same magnitude.

Problem 2: Charges, dipoles, and contour integrals (20 points)

Consider the complex electric field

$$f(z) = \frac{1}{2\pi} \left(\frac{2}{z-1} - \frac{1}{z+1} \right) + \frac{1}{(z-i)^2}.$$

All curves below are positively oriented.

1. Locate and state the strengths of the point charges. What does the term $1/(z-i)^2$ represent, and what is its net charge? (6 points)
2. Compute $\oint_{\gamma} f(z) dz$ for each of the following curves: (8 points)
 - $\gamma_1 : |z-1| = 1/4$;
 - $\gamma_2 : |z-i| = 1/4$;
 - $\gamma_3 : |z| = 2$.
3. For each curve, state the corresponding circulation and flux. (3 points)
4. Explain why the real and imaginary parts of f are harmonic away from the three singular points. (3 points)

Hints.

- A term $\frac{q}{2\pi(z-z_0)}$ represents a point charge q at z_0 .
- A second-order pole represents a dipole and has zero net charge.
- You may use

$$\oint_{|z-a|=r} \frac{dz}{z-a} = 2\pi i, \quad \oint_{|z-a|=r} \frac{dz}{(z-a)^m} = 0 \quad (m \geq 2).$$

- A term holomorphic on and inside a curve contributes zero to its integral.
- With $f = u - iv$,

$$\oint_{\gamma} f(z) dz = \text{circulation} + i \text{flux}.$$

Problem 3: Poincaré's inequality (25 points)

Let Ω be a finite union of grid squares contained in the strip $0 < x < L$. Suppose that v is continuous, piecewise C^1 , and zero on $\partial\Omega$. Extend v by zero outside Ω .

1. Explain why the extension is continuous and piecewise C^1 . (4 points)
2. Prove

$$\|v\|_{L^2(\Omega)} \leq L \|\nabla v\|_{L^2(\Omega)}, \quad \|F\|_{L^2(\Omega)}^2 := \iint_{\Omega} |F|^2 dA.$$

(15 points)

3. Suppose that every v_n vanishes on $\partial\Omega$ and that ∇v_n is Cauchy in L^2 . Prove that v_n is Cauchy in L^2 . Explain where this implication was used in Lecture 3. (6 points)

Hints.

- After extending by zero, fix y and write

$$v(x, y) = \int_0^x \partial_x v(s, y) ds.$$

- Apply Cauchy–Schwarz to the one-dimensional integral.
- Use $0 < x < L$, then integrate first in x and afterward in y .
- It is enough to bound $|\partial_x v|$ by $|\nabla v|$ after integration.
- For the last part, apply the inequality to $v_n - v_m$.

Problem 4: Using triangles instead of squares (35 points)

Let T be the triangle with corners

$$A = (0, 0), \quad B = (1, 0), \quad C = (0, 1).$$

A linear function on T has the form $a + bx + cy$. Its values at the three corners determine it everywhere on the triangle.

1. Find three linear functions ϕ_A, ϕ_B, ϕ_C . Each function must equal 1 at the corner named in its subscript and 0 at the other two corners. Compute the gradient of each function. (8 points)
2. For every pair $P, Q \in \{A, B, C\}$, calculate

$$A_{PQ} = \iint_T \nabla \phi_P \cdot \nabla \phi_Q dA.$$

Present the nine numbers as a 3×3 table. (10 points)

3. Now divide the unit square into four triangles by joining its centre

$$p_0 = \left(\frac{1}{2}, \frac{1}{2}\right)$$

to the four corners $p_1 = (0, 0)$, $p_2 = (1, 0)$, $p_3 = (1, 1)$, and $p_4 = (0, 1)$.

Let U be continuous on the square and linear on each triangle. Its values at the four corners are prescribed: $U(p_j) = g_j$. The only unknown value is $c = U(p_0)$.

Let ϕ_0 be the function that equals 1 at the centre, 0 at the four corners, and is linear on each triangle. The equation that determines c is

$$\iint_{(0,1)^2} \nabla U \cdot \nabla \phi_0 dA = 0.$$

Compute this integral one triangle at a time and prove that

$$c = \frac{g_1 + g_2 + g_3 + g_4}{4}.$$

Thus the value at the centre is the average of the four surrounding boundary values. (12 points)

4. Suppose that two continuous, piecewise-linear functions have the same boundary values and both satisfy

$$\iint_{(0,1)^2} \nabla U \cdot \nabla v \, dA = 0$$

for every continuous, piecewise-linear v that is zero on the boundary. Subtract the two functions and call the difference W . Choose $v = W$ and prove that $W = 0$. Thus the finite-element solution is unique. (5 points)

Hints and checks.

- To find each ϕ , substitute the three corner values into $a + bx + cy$ and solve for a, b, c .
- The gradients are constant, and the area of T is $1/2$.
- Your 3×3 table should be symmetric, and every row should sum to zero.
- On the bottom triangle with corners p_1, p_2, p_0 , the centre function is $\phi_0(x, y) = 2y$.
- Compute the bottom-triangle integral first. The other three calculations follow by rotation.
- For the last part, choosing $v = W$ gives

$$\iint_{(0,1)^2} |\nabla W|^2 \, dA = 0.$$

What does this say about W on a connected square? Then use its boundary values.

Optional bonus: FreeFEM++ (up to 15 points)

Submit your `.edp` source file, the requested figures, and brief answers to the interpretation questions. Begin early enough to resolve installation problems before the due date.

Part A: Equilibrium temperature in a house (up to 5 points)

Let $\Omega = (0, 4) \times (0, 3)$. A radiator on the part $0.5 < x < 1.5$ of the bottom boundary is held at 40° . A window on the part $1.5 < x < 2.5$ of the top boundary is held at 5° . The remainder of the boundary is held at 15° . There are no heat sources in the interior.

1. Construct and display a triangular mesh with separate labels for the radiator, window, and walls.
2. Solve $\Delta T = 0$ using continuous P_1 elements and plot the temperature with visible isotherms.
3. Report the numerical minimum and maximum. Explain how the maximum principle checks your result.

Hints.

- Split the top and bottom borders where the boundary values change.
- Plot the mesh before solving; incorrect boundary labels are a common source of errors.
- The essential structure is

```
fespace Vh(Th,P1);
Vh T,v;
problem heat(T,v)
  = int2d(Th)(dx(T)*dx(v)+dy(T)*dy(v))
  + on(/* wall labels */,T=15)
  + on(/* window label */,T=5)
  + on(/* radiator label */,T=40);
```

- Check that $5 \leq T \leq 40$ up to numerical roundoff.

Part B: WiFi propagation in a flat (up to 10 additional points)

Use the official FreeFEM++ tutorial *WiFi Propagation*. The tutorial models a complex electric field E by

$$\Delta E + \frac{k^2}{n^2} E = 0.$$

Here n is the refractive index. It equals 1 in air and is complex inside the walls: its real part controls reflection and its imaginary part controls absorption. The WiFi hotspot is represented by a small circular boundary on which $E = 1$.

1. Run the official example. Submit one picture of its mesh and one signal-strength plot. (2 points)
2. In a short paragraph, identify the outer boundary, the two walls, and the circular hotspot in the code. Explain what the line `on(2,v=1)` does. (2 points)
3. The FreeFEM++ problem contains the integral

$$\iint_{\Omega} \left(\frac{k^2}{n^2} E w - \nabla E \cdot \nabla w \right) dA.$$

Starting from the Helmholtz equation above, explain why this is the expression used in the program. (2 points)

4. Run the example for at least three hotspot positions. State which position gives the best coverage and explain what criterion you used. Include the three plots. (2 points)
5. Increase or decrease the absorption in the walls, leaving everything else unchanged. Include before-and-after plots and explain the difference. (2 points)

Hints and cautions.

- Start from the official script; you are not expected to retype the geometry from scratch.
- For the third task, multiply the equation by w and integrate the ΔE term by parts. The test function is zero on the hotspot, while the other boundaries carry the natural zero-normal-derivative condition.
- The function `wall()` distinguishes air from wall material.
- The functions `wallreflexion` and `wallabsorption` determine the complex refractive index inside the walls.
- The hotspot boundary has label 2, which is why `on(2,v=1)` prescribes its field value.

- The displayed quantity is the logarithm of the signal power: $\log(\text{real}(v)^2 + \text{imag}(v)^2)$.
- For a fair comparison, keep the mesh, random seed, frequency, and all unchanged parameters fixed.
- State your coverage criterion before comparing positions. For example, compare the size of the low-signal regions in the flat, or compute the same summary value for every position.